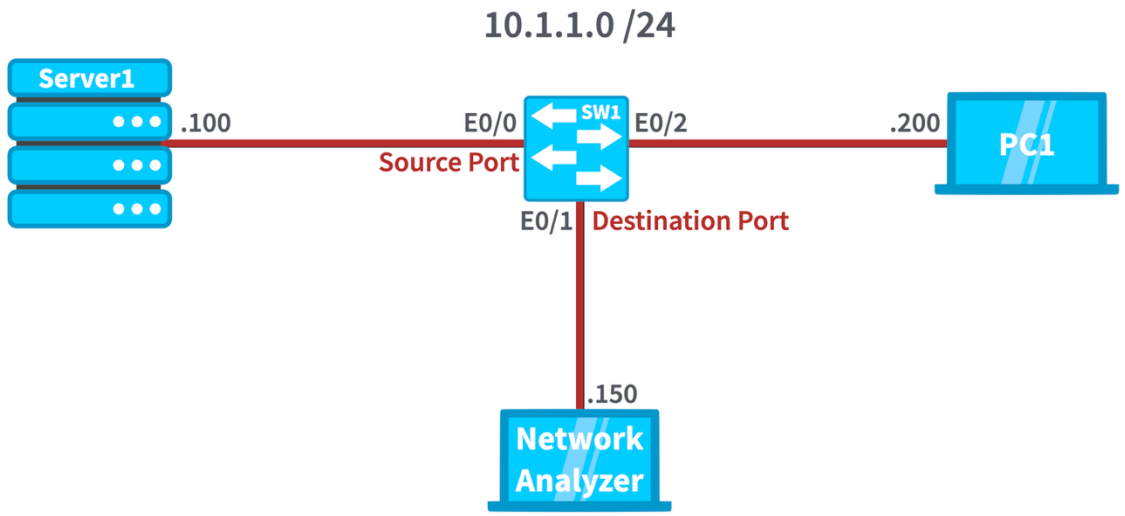
Local SPAN Lab

Topology



Initial Configuration Commands

SERVER1 (Server Node in CML):

Paste the following under the node's CONFIG tab in CML:

```
# this is a shell script which will be sourced at boot
hostname SERVER1
# configurable user account
USERNAME=cisco
PASSWORD=cisco
sudo hostname SERVER1
sudo ifconfig eth0 10.1.1.100 netmask 255.255.255.0 up
sudo route add default gw 10.1.1.1
```

PC1 (Desktop Node in CML):

Paste the following under the node's CONFIG tab in CML:

```
# this is a shell script which will be sourced at boot
hostname PC1
# configurable user account
USERNAME=cisco
PASSWORD=cisco
sudo hostname PC1
```

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```
sudo ifconfig eth0 10.1.1.200 netmask 255.255.255.0 up
sudo route add default gw 10.1.1.1
```

NETWORK-ANALYZER (Desktop Node in CML):

Paste the following under the node's CONFIG tab in CML:

```
# this is a shell script which will be sourced at boot
hostname NETWORK-ANALYZER
# configurable user account
USERNAME=cisco
PASSWORD=cisco
sudo hostname NETWORK-ANALYZER
sudo ifconfig eth0 10.1.1.150 netmask 255.255.255.0 up
sudo route add default gw 10.1.1.1
```

SW1:

```
enable
conf t
host SW1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 4
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
end
copy run star
```

Lab Task

1. Use the **`sudo tcpdump icmp`** command on the NETWORK-ANALYZER to monitor Ping traffic from PC1 to SERVER1 (10.1.1.100).
2. On PC1, issue the **`ping 10.1.1.100`** command to continuously Ping SERVER1.
3. Notice that no traffic is captured on the NETWORK-ANALYZER.
4. Configure Local SPAN on SW1 to mirror inbound and outbound traffic on E0/0 and send those mirrored packets out of a destination interface of E0/1.
5. Notice that the NETWORK-ANALYZER is capturing ICMP traffic.

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Solution

NETWORK-ANALYZER:

```
NETWORK-ANALYZER:~$ sudo tcpdump icmp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

PC1:

```
PC1:~$ ping 10.1.1.100
PING 10.1.1.100 (10.1.1.100): 56 data bytes
64 bytes from 10.1.1.100: seq=0 ttl=42 time=29.516 ms
```

NETWORK-ANALYZER:

[NOTICE NO TRAFFIC IS BEING CAPTURED]

SW1:

```
SW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#monitor session 1 source interface e0/0
SW1(config)#monitor session 1 destination interface e0/1
SW1(config)#end
SW1#show monitor session 1
Session 1
---------
Type                    : Local Session
Source Ports            :
    Both                : Et0/0
Destination Ports       : Et0/1
    Encapsulation       : Native


SW1#
```

NETWORK-ANALYZER:

```
18:59:19.829465 IP 10.1.1.200 > 10.1.1.100: ICMP echo request, id 1, seq 724,
length 64
18:59:19.830234 IP 10.1.1.100 > 10.1.1.200: ICMP echo reply, id 1, seq 724,
length 64
18:59:20.829752 IP 10.1.1.200 > 10.1.1.100: ICMP echo request, id 1, seq 725,
length 64
18:59:20.830368 IP 10.1.1.100 > 10.1.1.200: ICMP echo reply, id 1, seq 725,
length 64
```

[NOTICE ICMP TRAFFIC IS BEING CAPTURED]

Now, let's delete our existing monitoring session and set the source to VLAN 1 rather than a specific port or ports.

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SW1:

```
SW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#no monitor session 1
SW1(config)#monitor session 2 source vlan 1
SW1(config)#monitor session 2 destination interface e0/1
SW1(config)#end
SW1#show monitor session 2
Session 2
---------
Type                    : Local Session
Source VLANs            :
    Both                : 1
Destination Ports       : Et0/1
    Encapsulation       : Native


SW1#
```

NETWORK-ANALYZER:

```
19:05:39.035495 IP 10.1.1.200 > 10.1.1.100: ICMP echo request, id 1, seq
1103, length 64
19:05:39.035495 IP 10.1.1.200 > 10.1.1.100: ICMP echo request, id 1, seq
1103, length 64
19:05:39.036592 IP 10.1.1.100 > 10.1.1.200: ICMP echo reply, id 1, seq 1103,
length 64
```

[NOTICE ICMP TRAFFIC IS STILL BEING CAPTURED]

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